

Suman Jash

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SUMMARY

Computer Science Engineering student who designs, builds, and deploys full-stack, backend, and machine learning systems using Python, Flask, PHP, SQL, Docker, and GitHub Actions. Comfortable owning system design, security architecture, and model evaluation end-to-end; uses AI coding assistants the way a professional engineer does – to move faster, not to skip understanding. Codeforces Pupil and CodeChef 3-Star with 600+ problems solved.

EDUCATION

Narula Institute of Technology

B.Tech in Computer Science and Engineering, CGPA: 8.96/10

2023 – 2027

Kolkata, West Bengal

Arambagh Vivekananda Academy

CBSE Class XII – 85.8% | CBSE Class X – 94.4%

2021 – 2023

Arambagh, Hooghly

PROJECTS

- **CourierAI – ML-Powered Delivery Time Prediction Platform** | *Python, Flask, scikit-learn, SQLite, Docker*
GitHub | Live Demo
 - Trained a HistGradientBoostingRegressor on 49,639 real DTDC courier records via randomized search over 150/960 configurations with 5-fold CV, reaching 0.5361-day MAE and 0.7466 R^2 (CV std ± 0.0056 , train/test ratio 0.973 – no overfitting) at $\sim 5.8\mu s$ inference per prediction.
 - Built an experiment harness benchmarking 5 base models (Random Forest, XGBoost, CatBoost, SVR/SVC, MLP) against voting and stacking ensembles across up to 17 combinations, for both regression and classification.
 - Designed a secured REST API (`/api/predict`) with SHA-256-hashed and Fernet-encrypted per-user API keys, tiered rate limiting (30/min, 1000/day), and plan-based prediction quotas enforced with HTTP 402 responses.
 - Implemented bcrypt password hashing, CSRF protection, and security headers; built a standalone admin console covering users, plans, analytics, and live system/ML health.
 - Containerized with Docker and Docker Compose, added GitHub Actions CI running pytest, and deployed to Render behind Gunicorn.
- **AeroBook – Airline Reservation & Flight Operations Platform** | *PHP 8, MySQL, Bootstrap 5, JavaScript, Docker*
GitHub | Live Demo
 - Designed a normalized 24-table MySQL schema (FK constraints, 13 indexes) powering 24 passenger-facing pages and a 16-page admin Operations Center spanning flight search, booking, check-in, and account management.
 - Built a Smart Fare Engine that discovers connecting itineraries (90-minute to 8-hour layovers, single connection) in 3 database queries per search, scoring each on price, duration, layover length, and stop count to surface Best Value/Cheapest/Fastest options.
 - Built an interactive 2D seat map with multi-passenger booking (1-6 passengers), baggage/meal add-ons, promo code discounts, and QR-coded e-tickets with calendar export.
 - Built an admin analytics suite covering revenue, occupancy, and route reports with CSV exports, a searchable audit trail, and automated data-quality diagnostics; integrated live flight/airport data sync from the AviationStack API.
 - Hardened the app with prepared statements, CSRF protection, bcrypt hashing, and login lockout tracking; wrote a dependency-free smoke-test suite and shipped a containerized deployment with Docker.

TECHNICAL SKILLS

Languages: C/C++, Python, PHP, Java, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: Flask, scikit-learn, Pandas, NumPy, XGBoost, CatBoost, Bootstrap 5

Databases: MySQL, SQLite

Developer Tools: Git, GitHub, GitHub Actions, Docker, Docker Compose, pytest, VS Code, XAMPP

Deployment: Render, Vercel, InfinityFree

Core Concepts: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS, Machine Learning, REST APIs, Authentication, Web Security, CI/CD

ACHIEVEMENTS

- Secured 2nd rank in *Code It*, a college-level duo-format coding competition with 40 teams competing in the final round.
- Co-designed a School ERP System concept for Smart India Hackathon (SIH); owned system design, QA/testing, and delivered the final presentation after the idea was selected for the college-level demonstration round.
- Codeforces Pupil (max rating 1327) | CodeChef 3-Star (max rating 1628) | 600+ problems solved across LeetCode, CodeChef, and Codeforces.

CERTIFICATIONS

- Programming in Java (Elite) – NPTEL, Score: 94/100 | Cloud Computing – NPTEL, Score: 92/100
- Generative AI & Python Programming (30 hrs each) – Ardent Computech Pvt. Ltd.
- Prompt Engineering for AI & AI Automation and Deployment – Eduskill (Virtual Internship)